Guided Step-by-Step Demo: Combining Autoscaling and Load Balancing in Google Cloud

Overview & Objectives

In this production-tier integration demo, you will combine a Regional Managed Instance Group (MIG) equipped with an Autoscaler and front it with a Global External Application Load Balancer. This architecture is the foundational three-tier pattern for highly scalable, resilient web applications in Google Cloud.

By the end of this demo, you will be able to:

1. Deploy an auto-scaling regional Managed Instance Group.
2. Hook the regional MIG as a backend service behind a global Application Load Balancer.
3. Test end-to-end traffic distribution and verify automatic scaling under load.

Task 1: Create an Instance Template & Regional MIG with Autoscaling

1. Navigate to **Compute Engine > Instance templates** and click **Create instance template**:

    - **Name:** `prod-web-template`
    - **Machine type:** `e2-micro`
    - **Boot disk:** Ubuntu 24.04 LTS
    - **Startup script:**

      ```
      #!/bin/bash

      apt-get update

      apt-get install -y apache2 stress

      echo "Auto-scaled Backend Instance: $(hostname)" > /var/www/html/index.html

      systemctl restart apache2
      ```

- Click **Create**.

2. Navigate to **Compute Engine > Instance groups** and click **Create instance group**:

   - **Name:** `prod-regional-mig`
   - **Instance group type:** Managed instance group
   - **Instance template:** `prod-web-template`
   - **Location:** Regional (select `us-central1` across multiple zones).
   - **Autoscaling:** Configure autoscaling:
     - **Metric:** CPU utilization (Target: `50%`)
     - **Min instances:** `2`
     - **Max instances:** `5`
   - Click **Create**.

Task 2: Create Firewall Rules for Health Checks

1. Navigate to **VPC network > Firewall** and click **Create firewall rule**:
   - **Name:** `allow-lb-probes`
   - **Network:** `default`
   - **Source IP ranges:** `35.191.0.0/16` and `130.211.0.0/22`
   - **Protocols and ports:** TCP `80`
   - Click **Create**.

Task 3: Front the MIG with a Global External Application Load Balancer

1. Open the **Load balancing** page in the Google Cloud Console and click **Create load balancer**.
2. Select **Application Load Balancer (HTTP/HTTPS)** -> **Start configuration** -> **From Internet to my VMs** (External) -> **Global external Application Load Balancer**.
3. Name it `prod-frontend-alb`.
4. **Backend Configuration:**
   - Create a backend service named `prod-backend-service`.
   - Backend type: **Instance group** -> Select `prod-regional-mig`.
   - Port number: `80`.
   - Create a health check: HTTP on port 80, path `/`.
5. **Frontend Configuration:**
   - Protocol: HTTP (IPv4)

    - IP address: Ephemeral (Global Anycast IP).
6. Click **Create**.

Task 4: End-to-End Testing & Verification

1. Once the load balancer is active, copy its global IP address.
2. Run a continuous curl loop from your local terminal or Cloud Shell to watch traffic get distributed across instances:

   while true; do curl -s http://<LOAD_BALANCER_IP>; sleep 1; done

3. **Simulate Traffic Load:** SSH into one of the backend VMs and run `stress --cpu 4 --timeout 120`.
4. Observe how the Application Load Balancer continues routing traffic smoothly while the MIG automatically spins up new instances to handle the increased CPU load!